

Department of Computer Science and Engineering

Intelligent Meeting Memory Assistant for Executive Management

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Guide:

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Agenda

- **Problem Statement**
- **Objective & Scope of the Proposed Project**
- **Hardware & Software to be used**
- **What contribution to society would the project make?**
- **References**

Introduction

- **What:**

MeetRecall is an AI-based meeting memory assistant designed for executives and the people with neurodegenerative disorders in their early stages, age-related cognitive decline to remember individuals and conversation context from previous meetings.

- **Why:**

High-level officers meet many people daily and often forget identities, discussion points, and follow-up requirements, leading to communication gaps.

- **How:**

The system captures a person's image, records meeting audio, converts speech to text, generates an NLP-based summary, stores only the summary securely, and retrieves it during future meetings using facial recognition

Problem Statement

Existing System:

- Executives manually record meeting notes.
- CRM tools require manual search for history.
- Meeting AI tools only summarize, not contextually recall.

Limitations:

- Speech clarity
- High cognitive load on professionals.

Proposed System:

- Face-based identification of individuals.
- Automatic transcription and AI-generated summaries.
- Instant retrieval of past meeting context when the same person is detected.

Project Objectives

1. **To design a facial recognition-based identification system** that retrieves previously stored meeting summaries when the same individual meets the executive again.
2. **To develop an automatic speech-to-text conversion module** that converts recorded meeting audio into text accurately.
3. **To implement an NLP-based summarization system** that generates concise and editable meeting summaries.
4. **To ensure privacy and security** by deleting raw audio data immediately after processing and storing only summarized information.
5. **To provide administrative control features** such as editing summaries and deleting meeting records when no further interaction is required.

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Hardware Requirements:

- Smartphone with Camera
- Microphone
- Local/Cloud Server

Software Requirements:

- Python
- Node.js / FastAPI
- React Native / Flutter
- MongoDB

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Development Tools:

- VS Code
- Google Colab (Model Fine-tuning)
- Postman (API Testing)

Technologies Used:

- FaceNet / ArcFace (Pretrained Image Model)
- Whisper / Google STT (Pretrained Audio Model)
- T5-Small (Fine-tuned NLP Model)
- Cosine Similarity Matching
- REST APIs

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Thank You